# Selection: IF, ELSE IF and nested IF

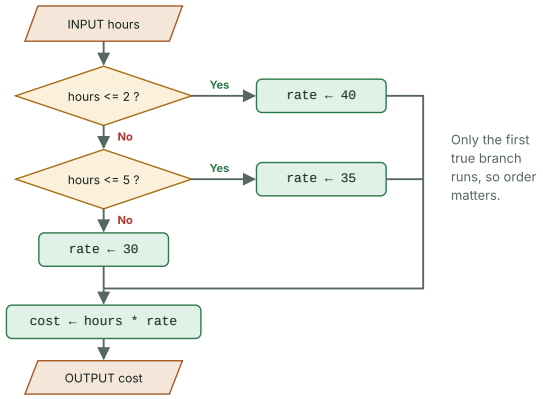
D15 FACT FILE

**Specification: students should be able to**

- Controlling the flow of a program through sequence, selection and iteration (selection)

CCEA fact file: Programming Constructs 3

**Key words:** sequence, selection, condition, IF, ELSE, ELSE IF / else if, nested IF, branch




Flowchart of an IF, ELSE IF, ELSE price calculation

## Selection

Selection lets a program take different paths. An IF statement runs a block only when its condition is True; IF–ELSE gives an alternative path; ELSE IF (else if in C#) adds further alternatives that are checked in order, so only the first matching branch runs. A nested IF places one decision inside another. Conditions use relational and Boolean operators.

## T Key terms from the mark schemes

**Sequence**
Statements executed one after another in order.
Credited: U4 2019 Q4

**Selection**
A condition decides which statements run (IF statements).
Credited: U4 2019 Q4, 2025 Q1

## C C# example

IF and IF–ELSE

```
Console.Write("Mark: ");
int mark = Convert.ToInt32(Console.ReadLine());
if (mark >= 50)
{
    Console.WriteLine("Pass");
}
else
{
    Console.WriteLine("Fail");
}
```

### Exam focus

Read every bullet point – they map to the mark scheme. Use the variable names supplied, use only one condition per IF if told to, include the calculation (e.g. total ← nights * hireCost using variable names, not numbers) and the output statement. Show where each IF ends.

| | | |
|---|---|---|
| Unit 4 2025 Q6 | [25] | Tent hire algorithm; Boolean; constants; validation true/false; Y/N validation loop |
| Unit 4 2025 Q3 | [15] | Musician hire algorithm; variables and data types; IDE aids; syntax vs logic |
| Unit 4 2025 Q1 | [5] | String type, compound AND, IF output, NOT/AND table, bubble sort |
| Unit 4 2024 Q3 | [17] | Travel insurance cost algorithm; validation; real data type; WHILE range-check loop |
| Unit 4 2024 Q1 | [4] | Dry-run a nested IF (shape calculator) |

## ? Check yourself

1. Describe the difference between sequence and selection. [2]
2. What is output when x = 7? IF x > 10 THEN OUTPUT 'A' ELSE IF x > 5 THEN OUTPUT 'B' ELSE OUTPUT 'C' END IF [1]
3. Write pseudocode that inputs a number of tickets and outputs the total: fewer than 5 tickets cost £8 each, 5 or more cost £6 each. [4]

**Answers:** 1. Sequence: statements run one after another in order (1) / selection: the program chooses which statements to run depending on whether a condition is True (1)  2. B (1)  3. INPUT tickets

```
IF tickets < 5 THEN
   price ← 8
ELSE
   price ← 6
END IF
total ← tickets * price
OUTPUT total
```

Marks: input number of tickets (1) / IF tickets < 5 gives £8 (1) / ELSE (or IF tickets >= 5) gives £6 (1) / total calculated with variables and output (1)